

1. Check your Ubuntu version

Open **VS Code** → **Terminal** → **New Terminal**.

Run:

```
lsb_release -a
```

You should see something similar to:

```
Distributor ID: Ubuntu
Description:   Ubuntu 24.04 LTS
Release:      24.04
Codename:     noble
```

You can also check whether MySQL is already installed:

```
mysql --version
```

If you get:

Command 'mysql' not found

that's fine. We will install it.

2. Update Ubuntu

In the VS Code terminal:

```
sudo apt update
```

Then:

```
sudo apt upgrade -y
```

Enter your Ubuntu password when requested.

3. Install MySQL Server

For a normal Ubuntu setup, install MySQL using:

```
sudo apt install mysql-server -y
```

Ubuntu will download and install MySQL Server and its required components.

MySQL's current documentation recommends installing MySQL through the APT repository on Ubuntu/Debian systems.

4. Check whether MySQL is running

Run:

```
sudo systemctl status mysql
```

You want to see:

Active: active (running)

For example:

- mysql.service - MySQL Community Server
Loaded: loaded
Active: active (running)

If it isn't running, execute:

```
sudo systemctl start mysql
```

Then check again:

```
sudo systemctl status mysql
```

MySQL documents `systemctl status mysql`, `start`, `stop`, and `restart` as the standard way to manage the service on systemd-based Ubuntu systems.

5. Make MySQL start automatically

Run:

```
sudo systemctl enable mysql
```

This means MySQL will automatically start when Ubuntu boots.

Verify:

```
sudo systemctl is-enabled mysql
```

Expected:

```
enabled
```

6. Test MySQL from the terminal

Try:

```
sudo mysql
```

You should enter the MySQL console:

Welcome to the MySQL monitor.

```
mysql>
```

Now execute:

```
SELECT VERSION();
```

You should get something similar to:

```
+-----+  
| VERSION() |  
+-----+  
| 8.x.x   |  
+-----+
```

Test another command:

```
SHOW DATABASES;
```

You should see databases such as:

```
information_schema
```

```
mysql
performance_schema
sys
```

Exit:

```
exit;
```

7. Create a MySQL user for VS Code

I recommend **not using root** from VS Code.

Instead, create a separate development user.

First enter MySQL:

```
sudo mysql
```

Then:

```
CREATE USER 'developer'@'localhost'
IDENTIFIED BY 'Developer@123';
```

Give the user privileges:

```
GRANT ALL PRIVILEGES ON *.* TO 'developer'@'localhost';
```

Apply:

```
FLUSH PRIVILEGES;
```

Check the user:

```
SELECT user, host
FROM mysql.user;
```

You should see:

```
developer  localhost
```

Exit:

```
exit;
```

For a real production environment, use a stronger password and grant only the privileges the application actually needs. For a training environment, the above is convenient.

8. Test the new user

Now test:

```
mysql -u developer -p
```

It will ask:

Enter password:

Enter:

Developer@123

You should get:

```
mysql>
```

Test:

```
SELECT USER();
```

Expected:

```
developer@localhost
```

Then:

```
exit;
```

At this point **MySQL itself is working correctly.**

9. Install VS Code

If VS Code is already installed, skip this section.

You can check:

```
code --version
```

If it displays a version, VS Code is installed.

If not, install VS Code from the official Microsoft site:

[Download Visual Studio Code](#)

10. Open VS Code

Start VS Code:

```
code .
```

Or open it from the Ubuntu applications menu.

You should see:

```
EXPLORER
SEARCH
SOURCE CONTROL
RUN AND DEBUG
EXTENSIONS
```

Click:

Extensions

or press:

```
Ctrl + Shift + X
```

11. Install a MySQL extension

In the Extensions search box, search:

```
MySQL
```

You will see several MySQL extensions.

For a training environment, choose a well-maintained MySQL database extension that supports:

- MySQL connections
- SQL execution
- Database explorer
- Tables
- Views
- Stored procedures
- SQL files

Alternatively, Oracle provides **MySQL Shell for VS Code**, which integrates MySQL development functionality into VS Code.

MySQL Shell for VS Code

Install the extension and reload VS Code if prompted.

12. Create your first SQL file

Create a folder for your SQL training.

For example:

mysql-training

Open it in VS Code.

Create:

01_basics.sql

Put this inside:

```
SELECT VERSION();
```

```
SHOW DATABASES;
```

Save the file.

13. Connect VS Code to MySQL

Open the MySQL extension from the VS Code sidebar.

Choose something similar to:

New Connection

Enter:

Host: localhost

Port: 3306

Username: developer

Password: Developer@123

Database can initially be left empty.

Your connection is essentially:

Host localhost

Port 3306

User developer

Password Developer@123

Then click:

Connect

14. Check the connection

Once connected, VS Code should show something similar to:

MYSQL

```
|
├── localhost:3306
│   ├── Databases
│   │   ├── information_schema
│   │   ├── mysql
│   │   ├── performance_schema
│   │   └── sys
```

If you can see the databases, **VS Code is successfully connected to MySQL.**

15. Create your training database

Let's create the database you'll use for your SQL training.

Open a SQL file:

```
CREATE DATABASE smartdb;
```

Execute it.

Then:

```
SHOW DATABASES;
```

You should now see:

```
smartdb
```

16. Select the database

Run:

```
USE smartdb;
```

Then:

```
SELECT DATABASE();
```

Expected:

```
+-----+  
| DATABASE() |  
+-----+  
| smartdb   |  
+-----+
```

17. Create your first table

Let's create a simple customer table:

```
CREATE TABLE customers
(
  customer_id INT PRIMARY KEY AUTO_INCREMENT,
  first_name VARCHAR(50) NOT NULL,
  last_name VARCHAR(50),
  email VARCHAR(100) UNIQUE,
  city VARCHAR(50),
  registration_date DATE DEFAULT (CURRENT_DATE)
);
```

Check it:

```
SHOW TABLES;
```

You should get:

```
customers
```

18. Insert data

Run:

```
INSERT INTO customers
(
  first_name,
  last_name,
  email,
  city
)
VALUES
('Arun', 'Kumar', 'arun@gmail.com', 'Chennai'),
('Priya', 'Sharma', 'priya@gmail.com', 'Bangalore'),
('Rahul', 'Verma', 'rahul@gmail.com', 'Mumbai');
```

19. Query the table

Run:

```
SELECT *  
FROM customers;
```

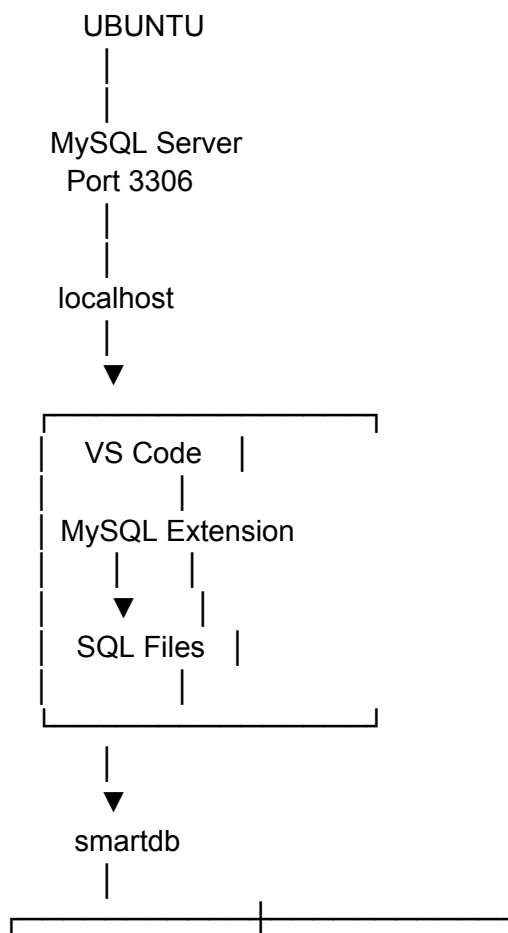
You should see something like:

customer_id	first_name	last_name	email	city
1	Arun	Kumar	arun@gmail.com	Chennai
2	Priya	Sharma	priya@gmail.com	Bangalore
3	Rahul	Verma	rahul@gmail.com	Mumbai

🎉 You have now successfully connected VS Code to MySQL and executed SQL.

20. Your final architecture

Your setup will look like this:



21. Useful commands to remember

Check MySQL version

```
mysql --version
```

Start MySQL

```
sudo systemctl start mysql
```

Stop MySQL

```
sudo systemctl stop mysql
```

Restart MySQL

```
sudo systemctl restart mysql
```

Check status

```
sudo systemctl status mysql
```

Login

```
mysql -u developer -p
```

Login as root

```
sudo mysql
```

Check databases

```
SHOW DATABASES;
```

Select database

```
USE smartdb;
```

Show tables

```
SHOW TABLES;
```

22. If VS Code cannot connect

If you get:

```
ECONNREFUSED 127.0.0.1:3306
```

check:

```
sudo systemctl status mysql
```

If MySQL isn't running:

```
sudo systemctl start mysql
```

Then check whether port **3306** is listening:

```
sudo ss -lntp | grep 3306
```

You should see something involving:

```
3306
```

Also test directly:

```
mysql -u developer -p
```

If this works from the terminal but VS Code doesn't, the problem is almost certainly the **VS Code connection/extension configuration**, not MySQL.

Recommended setup for your SQL training

Since you're using this for your **SmartMart SQL exercises**, I'd organize your VS Code project like this:

```
mysql-training/
|
|— 01_database/
|   |— create_database.sql
|
|— 02_tables/
|   |— customers.sql
|   |— products.sql
|   |— orders.sql
```

```
|
|   └─ order_items.sql
|
| └─ 03_insert/
|   └─ sample_data.sql
|
| └─ 04_select/
|   ├── select.sql
|   ├── where.sql
|   ├── order_by.sql
|   └─ group_by.sql
|
| └─ 05_joins/
|   ├── inner_join.sql
|   ├── left_join.sql
|   └─ self_join.sql
|
| └─ 06_subqueries/
|   ├── subquery.sql
|   └─ correlated_subquery.sql
|
| └─ 07_window_functions/
|   └─ window_functions.sql
```

This will give you a very good environment for demonstrating **SQL from basic SELECT all the way to joins, subqueries, correlated subqueries and window functions.**

Official MySQL documentation: [MySQL Getting Started](#) · [MySQL APT Repository Guide](#)